

REDIS Commands (Data Structures)

String

The simplest type in Redis, representing a single key-value pair where the value is a string. Strings can store any type of data, such as text, integers, or binary data like images or serialized objects.

Key Operations:

SET: Set the value of a key.

GET: Get the value of a key.

INCR, DECR: Increment/Decrement the value of a key by 1.

APPEND: Append to the value of a key.

Use Cases:

Caching single data items.

Storing basic counters, tokens, and session data.

List

Description: A list of ordered strings. Redis lists allow push and pop operations from both ends of the list (left and right), making them ideal for queue and stack implementations.

Key Operations:

LPUSH, RPUSH: Add elements to the left or right of the list.

LPOP, RPOP: Remove and get elements from the left or right.

LRANGE: Get a range of elements from the list.

BLPOP, BRPOP: Blocking variants of pop operations.

Use Cases:

Task queues, Time-ordered data streams, Messaging systems.

Set

Description: An unordered collection of unique strings. Redis sets allow you to perform common set operations like unions, intersections, and differences.

Key Operations:

SADD: Add elements to a set.

SREM: Remove elements from a set.

SMEMBERS: Get all the elements in a set.

SUNION, SINTER: Perform union and intersection on multiple sets.

Use Cases:

Storing unique elements, User group management.

Tagging systems.

Sorted Set (ZSET)

Description: Like a regular set but with an associated score for each member, which allows sorting. Members are unique, and the set is ordered by the score.

Key Operations:

ZADD: Add a member to a sorted set with a score.

ZRANGE, ZREVRANGE: Get elements in ascending or descending score order.

ZRANGEBYSCORE: Get elements within a specific score range.

ZINCRBY: Increment the score of a member.

Use Cases:

Leaderboards, Time-series data, Ranking systems.

Hash

Description: A hash map, or dictionary, that stores key-value pairs where both the key and value are strings. Hashes are ideal for representing objects and storing fields for specific entities.

Key Operations:

HSET: Set the value of a field in a hash.

HGET: Get the value of a field in a hash.

HGETALL: Get all fields and values from a hash.

HINCRBY: Increment a field value by a given integer.

Use Cases:

Storing user profiles, session information.

Caching database rows or objects.

String commands:

Command	Description	Example
SET	Set the value of a key.	SET student:1001:name "John Doe"
GET	Get the value of a key.	GET student:1001:name
MSET	Set multiple keys to multiple values.	MSET student:1001:name "John Doe" student:1001:age 21
MGET	Get the values of multiple keys.	MGET student:1001:name student:1001:age
INCR	Increment the integer value of a key by 1.	INCR student:1001:age
DECR	Decrement the integer value of a key by 1.	DECR student:1001:age
INCRBY	Increment the integer value of a key by the given amount.	INCRBY student:1001:score 10
DECRBY	Decrement the integer value of a key by the given amount.	DECRBY student:1001:score 5
INCRBYFLOAT	Increment the float value of a key by the given amount.	INCRBYFLOAT student:1001:gpa 0.2
APPEND	Append a value to a key.	APPEND student:1001:name " Jr."
GETSET	Set a new value and return the old value of a key.	GETSET student:1001:name "Johnny Doe"
STRLEN	Get the length of the value stored in a key.	STRLEN student:1001:name

Command	Description	Example
SETEX	Set the value of a key with expiration.	SETEX student:1001:session 600 "session_token_abc"
SETNX	Set the value of a key, only if the key does not exist.	SETNX student:1001:name "John Doe"
PSETEX	Set the value of a key with expiration in milliseconds.	PSETEX student:1001:session 1000 "session_token_xyz"
DEL	Delete one or more keys.	DEL student:1001:name
GETRANGE	Get a substring of the value stored at a key.	GETRANGE student:1001:name 0 3
SETRANGE	Overwrite part of a string starting at a specific position.	SETRANGE student:1001:name 5 "Smith"
BITCOUNT	Count the number of set bits (1s) in the string value stored at a key.	BITCOUNT student:1001:bitmap
BITOP	Perform bitwise operations between strings.	BITOP AND result_key key1 key2
EXISTS	Check if a key exists.	EXISTS student:1001:name

Keys based Commands:

Assume that following key-value pairs are already available in redis

student:101:name gopi

student:101:marks:math 98

student:101:marks:science 97

student:102:name krishnan

student:102:marks:math 87

student:102:marks:science 92

student:103:name Aadhav

student:103:marks:math 99

student:103:marks:science 95

class:10A:students 75

class:10B:students 56

attendance:101 96

attendance:102 48

Command	Description	Example Result
KEYS *	Matches all keys in the database.	student:101:name, student:101:marks:math, attendance:101
KEYS student:*	Matches all keys starting with student:.	student:101:name, student:101:marks:math, student:102:name
KEYS student:101:*	Matches all keys related to student 101.	student:101:name, student:101:marks:math, student:101:marks:science
KEYS *:marks:math	Matches all keys ending with :marks:math.	student:101:marks:math, student:102:marks:math
KEYS class:*:students	Matches all keys related to class students.	class:10A:students, class:10B:students

Command	Description	Example Result
KEYS student:*:marks:*	Matches all keys for student marks across subjects.	student:101:marks:math, student:101:marks:science, student:102:marks:math
KEYS student:10[1-3]:*	Matches keys for students 101, 102, and 103.	student:101:name, student:102:marks:math, student:103:marks:science
KEYS attendance:*	Matches all keys related to attendance records.	attendance:101, attendance:102
KEYS *:10A:*	Matches all keys containing :10A: (related to class 10A).	class:10A:students
KEYS student:???	Matches all keys that have exactly 3 characters after student:	No match (since keys have more than 3 characters after student:)

List commands:

Command	Description	Example
LPUSH	Add one or more elements to the beginning of a list.	LPUSH student:1001:courses "Math" "Science"
RPUSH	Add one or more elements to the end of a list.	RPUSH student:1001:courses "History" "English"
LSET	Set the value of an element in a list by its index.	LSET student:1001:courses 1 "Physics"
LINSERT	Insert an element before or after another element in a list.	LINSERT student:1001:courses BEFORE "Math" "Algebra"
LPOP	Remove and get the first element in a list.	LPOP student:1001:courses
RPOP	Remove and get the last element in a list.	RPOP student:1001:courses
LLEN	Get the length of a list.	LLEN student:1001:courses
LRANGE	Get a range of elements from a list.	LRANGE student:1001:courses 0 -1
LREM	Remove elements from a list.	LREM student:1001:courses 2 "Math"
LINDEX	Get an element from a list by its index.	LINDEX student:1001:courses 2
LTRIM	Trim a list to the specified range.	LTRIM student:1001:courses 0 2
LPUSHX	Add an element to the head of a list, only if the list exists.	LPUSHX student:1001:courses "Art"

Command	Description	Example
RPUSHX	Add an element to the tail of a list, only if the list exists.	RPUSHX student:1001:courses "Geography"
RPOPLPUSH	Remove the last element in a list, and push it to the head of another list.	RPOPLPUSH student:1001:courses student:1002:courses
BLPOP	Remove and get the first element in a list, blocking until one is available.	BLPOP student:1001:courses 0
BRPOP	Remove and get the last element in a list, blocking until one is available.	BRPOP student:1001:courses 0
BRPOPLPUSH	Remove the last element in a list and push it to another list, blocking if necessary.	BRPOPLPUSH student:1001:courses student:1002:courses 0

Set commands:

Command	Description	Example
SADD	Add one or more members to a set.	SADD student:1001:skills "Python" "Java"
SREM	Remove one or more members from a set.	SREM student:1001:skills "Java"
SMEMBERS	Get all the members in a set.	SMEMBERS student:1001:skills
SISMEMBER	Check if a member exists in a set.	SISMEMBER student:1001:skills "Python"
SCARD	Get the number of members in a set.	SCARD student:1001:skills
SPOP	Remove and return a random member from a set.	SPOP student:1001:skills
SRANDMEMBER	Get one or more random members from a set without removing them.	SRANDMEMBER student:1001:skills 2
SMOVE	Move a member from one set to another.	SMOVE student:1001:skills student:1002:skills "Python"
SUNION	Get the union of multiple sets.	SUNION student:1001:skills student:1002:skills
SUNIONSTORE	Store the union of multiple sets in a new set.	SUNIONSTORE student:all:skills student:1001:skills student:1002:skills

Command	Description	Example
SINTER	Get the intersection of multiple sets.	SINTER student:1001:skills student:1002:skills
SINTERSTORE	Store the intersection of multiple sets in a new set.	SINTERSTORE student:common:skills student:1001:skills student:1002:skills
SDIFF	Get the difference between sets.	SDIFF student:1001:skills student:1002:skills
SDIFFSTORE	Store the difference between sets in a new set.	SDIFFSTORE student:unique:skills student:1001:skills student:1002:skills

SADD students:math 101 102 103

SADD students:science 102 103 104

SADD students:english 101 104 105

Operation	Command Example	Result
Union	SUNION students:math students:science	101 102 103 104
Intersection	SINTER students:math students:science	102 103
Difference	SDIFF students:math students:science	101
Complex Diff	SDIFF (SUNION students:math students:science) students:english	102 103

Sorted Set (ZSET) commands:

Command	Description	Example
ZADD	Add one or more members with scores to a sorted set.	ZADD student:1001:scores 95 "Math" 88 "Science"
ZREM	Remove one or more members from a sorted set.	ZREM student:1001:scores "Math"
ZRANGE	Return a range of members in a sorted set, by index (ascending).	ZRANGE student:1001:scores 0 -1
ZREVRANGE	Return a range of members in a sorted set, by index (descending).	ZREVRANGE student:1001:scores 0 -1
ZRANGEBYSCORE	Return a range of members in a sorted set, by score (ascending).	ZRANGEBYSCORE student:1001:scores 80 95
ZREVRANGEBYSCORE	Return a range of members in a sorted set, by score (descending).	ZREVRANGEBYSCORE student:1001:scores 95 80
ZSCORE	Get the score associated with a member in a sorted set.	ZSCORE student:1001:scores "Math"
ZINCRBY	Increment the score of a member in a sorted set.	ZINCRBY student:1001:scores 5 "Math"
ZRANK	Determine the index of a member in a sorted set (ascending).	ZRANK student:1001:scores "Math"
ZREVRANK	Determine the index of a member in a sorted set (descending).	ZREVRANK student:1001:scores "Math"
ZCOUNT	Count the members in a sorted set with scores within a specific range.	ZCOUNT student:1001:scores 80 95

Command	Description	Example
ZCARD	Get the number of members in a sorted set.	ZCARD student:1001:scores
ZREM	Remove one or more members from a sorted set.	ZREM student:1001:scores "Math"
ZLEXCOUNT	Count the number of members in a sorted set between a lexicographical range.	ZLEXCOUNT student:1001:scores [A [Z
ZRANGEBYLEX	Return members in a sorted set between a lexicographical range.	ZRANGEBYLEX student:1001:scores [A [Z
ZREMRANGEBYRANK	Remove all members in a sorted set by their rank.	ZREMRANGEBYRANK student:1001:scores 0 2
ZREMRANGEBYSCORE	Remove all members in a sorted set with scores within a given range.	ZREMRANGEBYSCORE student:1001:scores 80 95
ZPOPMIN	Remove and return the member with the lowest score from a sorted set.	ZPOPMIN student:1001:scores
ZPOPMAX	Remove and return the member with the highest score from a sorted set.	ZPOPMAX student:1001:scores

ZADD students:math 90 101 85 102 80 103

ZADD students:science 88 102 95 103 78 104

ZADD students:science 88 102 95 103 78 104

Operation	Command Example	Example Result
Union	ZUNIONSTORE students:total 2 students:math students:science	103 175, 102 173, 104 78, 101 90

Operation	Command Example	Example Result
Intersection	ZINTERSTORE students:common 2 students:math students:science	103 175, 102 173
Difference	ZUNIONSTORE students:math_diff 1 students:math ZREM students:math_diff 102 103	101 90

Hash commands:

Command	Description	Example
HSET	Set the value of a field in a hash.	HSET student:1001 name "John Doe" age 21 major "CS"
HGET	Get the value of a field in a hash.	HGET student:1001 name
HGETALL	Get all fields and values in a hash.	HGETALL student:1001
HMSET	Set multiple fields in a hash (deprecated in favor of HSET).	HMSET student:1001 name "John Doe" age 21
HMGET	Get the values of multiple fields in a hash.	HMGET student:1001 name age
HDEL	Delete one or more fields from a hash.	HDEL student:1001 major
HEXISTS	Check if a field exists in a hash.	HEXISTS student:1001 age
HLEN	Get the number of fields in a hash.	HLEN student:1001
HINCRBY	Increment the integer value of a field by a given amount.	HINCRBY student:1001 age 1
HINCRBYFLOAT	Increment the float value of a field by a given amount.	HINCRBYFLOAT student:1001 gpa 0.5
HKEYS	Get all the fields (keys) in a hash.	HKEYS student:1001

Command	Description	Example
HVALS	Get all the values in a hash.	HVALS student:1001
HRANDFIELD	Get one or more random fields from a hash.	HRANDFIELD student:1001
HSTRLEN	Get the length of the value of a field in a hash.	HSTRLEN student:1001 name
HSCAN	Incrementally iterate over fields and values in a hash.	HSCAN student:1001 0 MATCH "name*"

Summary:

Data Structure	Insert Command	Show Command	Remove Command	Update Command
List	LPUSH student:1001:courses "Math" or RPUSH student:1001:courses "Science"	LRANGE student:1001:courses 0 -1	LPOP student:1001:courses or RPOP student:1001:courses or LREM student:1001:courses 1 "Math"	LSET student:1001:courses 1 "Physics"
Set	SADD student:1001:skills "Python"	SMEMBERS student:1001:skills	SREM student:1001:skills "Python"	N/A (Remove and reinsert)
Sorted Set	ZADD student:1001:scores 95 "Math"	ZRANGE student:1001:scores 0 -1 or ZREVRANGE student:1001:scores 0 -1	ZREM student:1001:scores "Math"	ZINCRBY student:1001:scores 5 "Math"

Hash	HSET student:1001 name "John Doe" age 21	HGETALL student:1001 or HMGET student:1001 name age	HDEL student:1001 age	HSET student:1001 age 22
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Redis administration commands:

Command	Description	Example
INFO	Get server information and statistics.	INFO
CONFIG GET	Get a specific Redis configuration.	CONFIG GET maxmemory
CONFIG SET	Set a Redis configuration parameter.	CONFIG SET maxmemory 256mb
CLIENT LIST	List all connected clients.	CLIENT LIST
CLIENT KILL	Kill a specific client connection.	CLIENT KILL 127.0.0.1:6379
FLUSHDB	Delete all keys in the current database.	FLUSHDB
FLUSHALL	Delete all keys in all databases.	FLUSHALL
SAVE	Synchronously save the dataset to disk.	SAVE
BGSAVE	Save the dataset to disk in the background.	BGSAVE
LASTSAVE	Get the UNIX timestamp of the last successful save.	LASTSAVE

Command	Description	Example
SHUTDOWN	Shutdown the Redis server.	SHUTDOWN
BGREWRITEAOF	Rewrites the Append-Only File in the background.	BGREWRITEAOF
DBSIZE	Get the number of keys in the currently selected database.	DBSIZE
CLIENT PAUSE	Temporarily suspend all clients.	CLIENT PAUSE 5000
SLOWLOG GET	View the Redis Slow Log.	SLOWLOG GET 10
SLOWLOG RESET	Clear the Slow Log.	SLOWLOG RESET
CLIENT REPLY	Control the replies sent by Redis to clients.	CLIENT REPLY ON
MONITOR	Monitor all commands processed by the server.	MONITOR
ROLE	Get the role of the Redis instance (master/slave).	ROLE
SLAVEOF	Configure replication to make the Redis instance a replica	SLAVEOF 192.168.1.100 6379
SLAVEOF NO ONE	Stop replication and make the instance a master again.	SLAVEOF NO ONE
DEBUG OBJECT	Get debugging information about a key.	DEBUG OBJECT student:1001
DEBUG SEGFAULT	Crash the Redis server (testing purpose only).	DEBUG SEGFAULT

Command	Description	Example
MEMORY USAGE	Get memory usage for a key.	MEMORY USAGE student:1001
MEMORY STATS	Get a summary of memory usage.	MEMORY STATS
MEMORY PURGE	Attempt to reclaim memory by purging volatile memory.	MEMORY PURGE
CLIENT ID	Get the ID of the current client connection.	CLIENT ID
CLIENT UNBLOCK	Unblock a blocked client.	CLIENT UNBLOCK 5
TIME	Get the current server time.	TIME
AUTH	Authenticate to the Redis server (if password-protected).	AUTH my_password
CLIENT SETNAME	Set a name for the current connection.	CLIENT SETNAME admin
REPLICAOF	Alias for SLAVEOF to configure replication.	REPLICAOF 192.168.1.100 6379